

6a Big-O notation

Friday, January 26, 2024

10:24 AM

$x \rightarrow \infty$ case

↖ "Big O"

$f = O(g)$ means asymptotically, f grows no faster than g

$f = o(g)$ means asymptotically, f grows slower than $\alpha \cdot g \quad \forall \alpha \in \mathbb{R}^+$

"little o" Formally: f, g positive real-valued functions ($f, g: \mathbb{R} \rightarrow \mathbb{R}^+$)

Equivalently,
 $\limsup_{x \rightarrow \infty} \frac{f(x)}{g(x)} < \infty$

order unimportant

$[f = O(g) \text{ means } \exists x_0 \geq 0, \exists \alpha \geq 0 \text{ st } \forall x > x_0, f(x) \leq \alpha g(x)]$

Equivalently,

$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 0$

$[f = o(g) \text{ means } \forall \alpha > 0, \exists x_0 \text{ st } \forall x > x_0, f(x) \leq \alpha g(x)]$

So... $f = o(g) \Rightarrow f = O(g)$
stronger

2026, do this as in-class exercise,
give a few possible definitions, i.e.,

1) true def

2) swap order of x_0, α (fine)

3) $\exists x_0 \text{ st } \forall \alpha > 0, \forall x > x_0, f(x) \leq \alpha g(x)$
[too strong: $\Rightarrow f = o$]

$f = \Omega(g)$ means $g = O(f)$

$f = \omega(g)$ means $g = o(f)$

$f = \Theta(g)$ means $f = O(g)$ and $g = O(f)$

$f \sim g$ is an even stronger version of this

$f = \tilde{O}(g)$ means $\exists k \in \mathbb{N}$ st. $f = O(g \log^k(g))$

This is all setup for "computer science" where we ask
about, say, runtime as a function of problem size
" $f(x)$ " " x ", $x \rightarrow \infty$

Ex: $h = O(h^2)$

$\log(h) = O(h)$, $h^2 \log(h) = O(h^3)$
 $= \tilde{O}(h^2)$

$x \rightarrow 0$ case

But analogous notation exists for "analysis" where " x " gets small

i.e. $f(h), h \rightarrow 0$

Ex: $h^2 = O(h)$ (as $h \rightarrow 0$)

NOT TRUE in $h \rightarrow \infty$ case